

Multiagent Reinforcement Learning (MARL): Foundations



Outline

- **Fundamentals of MARL**
- Types of MAS
- Why MARL is hard
- From MDPs to MARL
- CTDE



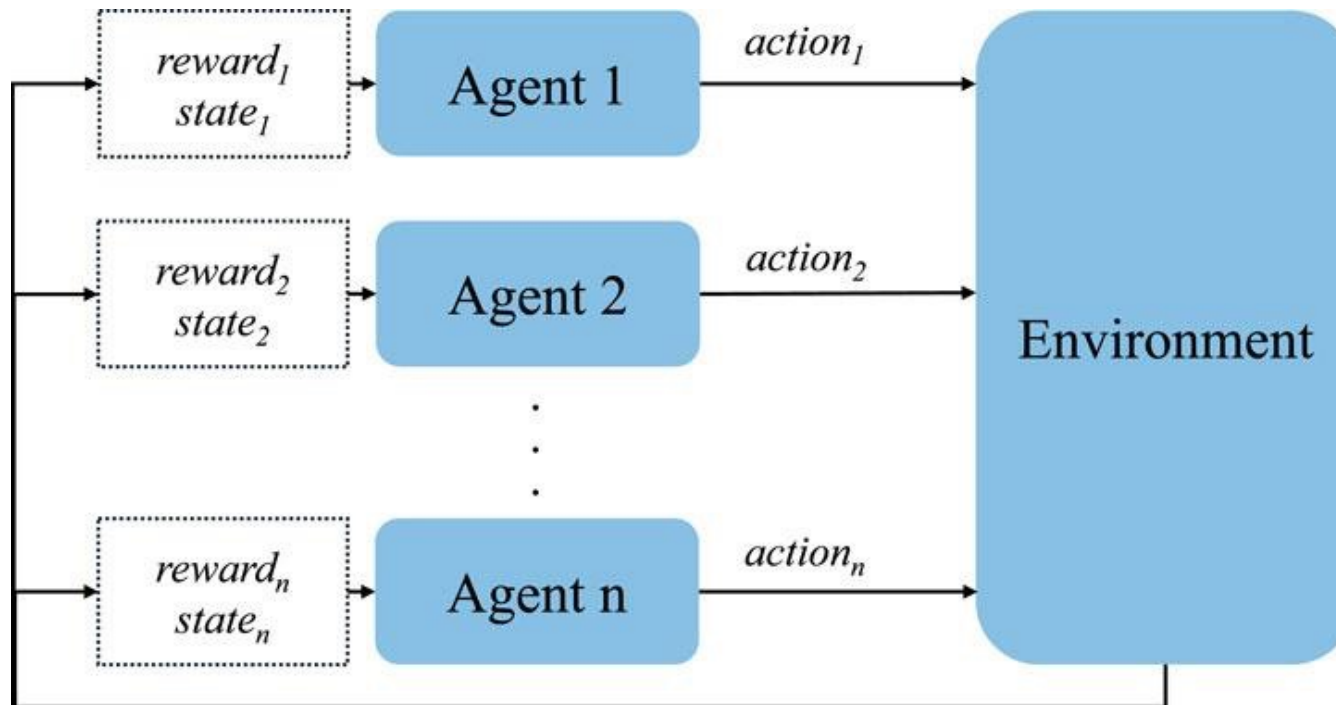
Why Multi-Agent Systems?

- **Multiple decision-makers** in real-world problems
- **Shared environment** interaction
- Examples: robotics, traffic, markets, games

What is an Agent?

- Agent's loop:
 - Observes environment
 - Takes actions
 - Receives rewards
- **Multiple agents may have different goals**

MARL Interaction Loop



- Agents choose actions simultaneously (joint action)
- Environment transitions
- Each agent receives observation and reward

Joint Action and Rewards

- Transition depends on the joint action of all agents
- Shared or individual rewards
- Possibly conflicting objectives

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Types of Multi-Agent Settings

- Cooperative (shared reward)
- Competitive (zero-sum)
- General-sum (mixed incentives)

Types of Multi-Agent Settings

- **Cooperative** (shared reward)
 - Example: A team of warehouse robots working together to pick and deliver orders. All robots benefit when an order is completed, so they can share the same reward signal.

Types of Multi-Agent Settings

- **Competitive** (zero-sum)
 - Example: A two-player board game such as chess, Go, or a competitive video game. If one player wins, the other loses, so one agent's reward is the negative of the other's.

Types of Multi-Agent Settings

- **General-sum** (mixed incentives)
 - Example: Autonomous driving is a very good example. Cars must cooperate in some sense to avoid collisions, but each car also has its own objective, such as minimizing travel time and driving smoothly. So their incentives are partly aligned and partly conflicting.

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Why MARL is Hard

- Non-stationarity
- Credit assignment
- Scalability
- Exploration challenges

Non-Stationarity

- Other agents update their policies during learning
- This makes the **environment non-stationary** from each agent's perspective

Non-Stationarity

- Example: At an intersection, one car may first learn to yield, then later learn to cross more aggressively, changing the learning problem faced by the other car
- Implication: **learning can be unstable or cyclic**

Credit Assignment

- A team may receive a reward, but it is **unclear which agent caused it**
- In MARL, it is hard to **determine which agents' actions contributed to success**

Credit Assignment

- Example: A soccer team scores a goal and all agents receive a positive reward, but it is unclear whether the goal was mainly due to the striker, the earlier pass, the defender recovering the ball, or the goalkeeper's previous save
- Implication: **wrong behaviors may be reinforced**

Scalability

- **Joint action space grows exponentially**
- $|A| = |A_1| \times |A_2| \times \dots \times |A_n|$
- Example: more agents $\rightarrow$ many more joint actions
- Implication: **slower and harder learning**

Exploration Challenges

- In MARL, good outcomes **may require several agents to discover compatible behaviors** at the same time
- Random exploration by one agent alone may not be enough
- Requires **coordinated exploration**

Exploration Challenges

- Example: Two warehouse robots must lift and carry a heavy box together. If only one robot tries to lift, nothing happens. The reward appears only when both robots coordinate their actions at the same time
- Implication: **good behaviors are hard to discover**

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From MDPs to MARL



MDP: single action

MARL: joint action

Leads to stochastic games

Stochastic Games

- Key features of Stochastic Game:
 - States, agents, actions
 - Transition depends on joint action
 - Each agent has its own reward

Partial Observability

- Agents may not see full state
- Receive local observations
- Leads to **Partially Observable Stochastic Games (POSG)**

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Centralized vs Decentralized in MARL

- In MARL, we distinguish between:
 - **Training** (how agents learn)
 - **Execution** (how agents act)
- These are **two different things**

Decentralized Execution

- Each agent acts **independently**
- Uses only its **own local observation**
- **No central controller** at decision time

Decentralized Execution

- **Example:**
 - Autonomous vehicles in traffic
 - Each car decides its actions based on its own sensors
- **Required in most real-world systems**

Centralized Training with Decentralized Execution (CTDE)

- **During training**, the system can use **global information**:
 - full state
 - all agents' actions
 - shared simulator
- **During execution**:
 - Each agent acts **independently**
- **Helps learning and coordination**

Centralized Training with Decentralized Execution (CTDE)

- **Example:** Warehouse robots
- **Training** → full map and all robots visible
- **Execution** → each robot uses only local sensors

Key Challenges

- Non-stationarity
- Partial observability
- Credit assignment
- Coordination vs competition

Thank You



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